

MNIST handwritten digit Classification

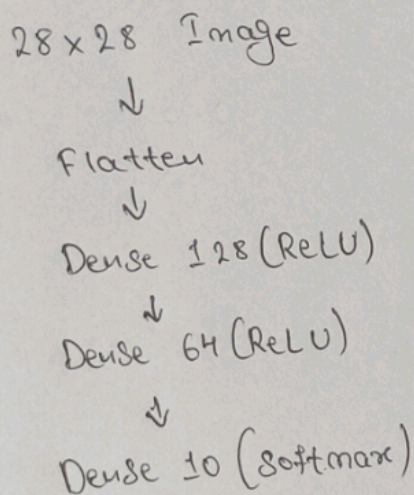
Introduction:

The aim of this assignment is to classify handwritten digits (0-9) using neural network built with TensorFlow/Keras.

Dataset:

MNIST contains 60,000 training images and 10,000 testing images. Each image is a 28×28 grayscale image. Pixel values were normalized from 0-255 to 0-1.

Model:



Training:

- *] Optimizer: Adam
- *] Loss: Sparse Categorical Crossentropy
- *] Epochs: 10
- *] Batch Size: 32

Result:

Test Accuracy: — %

Handwritten Image Testing

Image	Actual	Predicted
Image 1	0	5
Image 2	1	5
Image 3	2	5
Image 4	3	5
Image 5	4	5

Experiment : Dropout

A dropout rate of 0.2 was added after the hidden layers to reduce overfitting

<u>Model</u>	<u>Accuracy</u>
original	97.47%
Dropout	98.08%

Conclusion :

The neural network successfully classified MNIST handwritten digits with high accuracy. The model was also tested on five handwritten images. The dropout experiment showed how ~~can~~ changing the network architecture can affect model performance.